

Maintenance Schedule

Frac Pack

MTU Series 4000 12 V 4000 T94L

PFRPK12V4000-1A0

Oil & Gas

Application Group 4D

MS50093/02E



Power. Passion. Partnership.

Table 1: applicable for

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

1 Maintenance Schedule

1.1 Preface

MTU maintenance concept

The following applies in the case of engines which are subject to emissions regulations:

Emissions regulations prohibit alteration, removal or addition of any mechanical or electronic component or calibration that could affect the emissions characteristics of the engine. Maintenance, replacement, or repair of emissions control devices, sensors, cylinder heads and exhaust flaps, for example, can only be performed using MTU approved components or equivalents.

Failure to adhere to these guidelines could result in violation of legal requirements.

MTU Maintenance Schedules ensure the reliability and performance of MTU engines and must be adhered to during the full life of the engine. In the case of engines which are not subject to any emissions regulations, the maintenance intervals cited are to be regarded as guideline values.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high level of equipment availability.

The maintenance schedule is based on the load profile / load factor specified below. Special operating conditions and technical requirements may require additional maintenance work and/or modification of the maintenance intervals. The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. The limit which occurs first is applicable. In order to be authorized to carry out the individual maintenance jobs, maintenance personnel must have achieved a level of training and qualification appropriate to the complexity of the task in hand. The various Qualification Levels QL1 to QL4 reflect the levels of training offered in MTU courses and the contents of the tool kits required:

QL1: Operational monitoring and maintenance which can be carried out during a break in operation without disassembling the engine.

QL2: Component exchange (corrective only).

QL3: Maintenance work which requires partial disassembly of the engine.

QL4: Maintenance work which requires complete disassembly of the engine.

The maintenance schedule matrix normally finishes with extended component maintenance. Following this, maintenance work is to be continued at the intervals indicated.

The numbers stated in the list of jobs provide a reference to the scope of parts required.

Notes on maintenance

Specifications for fluids and lubricants, guideline values for their maintenance and change intervals and lists of recommended fluids and lubricants are contained in the MTU Fluids and Lubricants Specifications A001061 and in the fluids and lubricants specifications produced by the component manufacturers. They are therefore not included in the maintenance schedule (exception: deviations from the Fluids and Lubricants Specifications). All fluids and lubricants used must meet MTU specifications and be approved by the relevant component manufacturer.

Amongst other items, the operator/customer must carry out the following additional maintenance work:

- Protect components made of rubber or synthetic material from oil. Never treat them with organic detergents. Wipe with a dry cloth only.
- Fuel prefilter:
The maintenance interval depends on how dirty the fuel is. The paper inserts in fuel prefilters must be changed every two years at the latest (Task 9998).
- Battery:
Battery maintenance depends on the level of use and the ambient conditions. The battery manufacturer's instructions must be obeyed.

The relevant manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

This Maintenance Schedule may include components which are not included in the MTU supply scope; these may be disregarded.

Out-of-service periods

If the engine is to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Fluids and Lubricants Specifications, MTU Publication No. A001061.

Application Group

4D	Frac-operation (continuous operation with low load)
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Load factor

Load factor	<35%				
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Load profile

Load	90%	35%	20%			
Corresponding operating time	7%	77%	16%			

1.2 Maintenance schedule matrix

0-10,500 Operating hours

Item	Limit	Operating hours [h]																					
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500
Engine																							
Emergency air shut-off flaps	1 a																						
Engine oil filter	2 a																						
Centrifugal oil filter	2 a																						
Engine operation	-	X																					
Fuel filter	2 a			X		X		X		X		X		X		X		X		X		X	
Belt drive	2 a			X		X		X		X		X		X		X		X		X		X	
Primary fuel filter	2 a			X		X		X		X		X		X		X		X		X		X	
Valve gear	2 a							X						X						X			
Coolant filters	2 a							X						X						X			
Exhaust gas recirculation	2 a							X						X						X			
NOx sensor	6 a										X									X			
Lambda sensor	6 a										X									X			
Fuel injectors / injectors	9 a										X									X			
Engine governor	9 a										X									X			
Combustion chambers	6 a										X									X			
Engine mounts	9 a										X									X			
Turbochargers	18 a										X									X			
Exhaust gas recirculation	6 a										X									X			
Engine mounts	5 a										X									X			
Turbochargers	18 a																			X			
Component maintenance	18 a																			X			
Crankcase breathers	6 a																			X			
Fuel delivery pump	18 a																			X			
Battery charging generator	18 a																			X			
Cylinder heads	18 a																			X			
Hose lines	6 a																			X			
Extended component maintenance	18 a																			X			
Engine mounts	18 a																			X			
Transmission																							
Transmission operation	-	X																					
Gear oil filter	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Transmission mounts	-			X		X		X		X		X		X		X		X		X		X	
Screw connections	-			X		X		X		X		X		X		X		X		X		X	
Connectors/wiring harness	-			X		X		X		X		X		X		X		X		X		X	
Torsional resilient coupling	6 m					X				X				X				X				X	
Suction strainer (magnetic)	-																			X			
Transmission	-																			X			
Preheater / Jacket water heater																							
Preheater / Jacket water heater	1 a																						
Fuel pump	5 a																						
Circulation pump	-																						
w = weeks m = months a = years																							

Item	Limit	Operating hours [h]																
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000
Frac Pack																		
Exhaust silencer	-	X																
Frac Pack securing screws	-	X																
Frac Pack securing screws	-		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Air filters	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
w = weeks m = months a = years																		

11,000-18,000 Operating hours

Item	Limit	Operating hours [h]																
		11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500	15,000	15,500	16,000	16,500	17,000	17,500	18,000		
Engine																		
Emergency air shut-off flaps	1 a																	
Engine oil filter	2 a																	
Centrifugal oil filter	2 a																	
Engine operation	-																	
Fuel filter	2 a	X		X		X		X		X		X		X		X		
Belt drive	2 a	X		X		X		X		X		X		X		X		
Primary fuel filter	2 a	X		X		X		X		X		X		X		X		
Valve gear	2 a			X						X						X		
Coolant filters	2 a			X						X						X		
Exhaust gas recirculation	2 a			X						X						X		
NOx sensor	6 a						X									X		
Lambda sensor	6 a						X									X		
Fuel injectors / injectors	9 a						X									X		
Engine governor	9 a						X									X		
Combustion chambers	6 a						X									X		
Engine mounts	9 a						X									X		
Turbochargers	18 a						X									X		
Exhaust gas recirculation	6 a						X									X		
Engine mounts	5 a						X									X		
Turbochargers	18 a															X		
Component maintenance	18 a															X		
Crankcase breathers	6 a															X		
Fuel delivery pump	18 a															X		
Battery charging generator	18 a															X		
Cylinder heads	18 a															X		
Hose lines	6 a															X		
Extended component maintenance	18 a															X		
Engine mounts	18 a															X		
Transmission																		
Transmission operation	-																	
Gear oil filter	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		
Transmission mounts	-	X		X		X		X		X		X		X		X		
Screw connections	-	X		X		X		X		X		X		X		X		
w = weeks m = months a = years																		

Item	Limit	Operating hours [h]															
		11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500	15,000	15,500	16,000	16,500	17,000	17,500	18,000	
Connectors/wiring harness	-	X		X		X		X		X		X		X		X	
Torsional resilient coupling	6 m			X				X				X				X	
Suction strainer (magnetic)	-															X	
Transmission	-															X	
Preheater / Jacket water heater																	
Preheater / Jacket water heater	1 a																
Fuel pump	5 a																
Circulation pump	-															X	
Frac Pack																	
Exhaust silencer	-																
Frac Pack securing screws	-																
Frac Pack securing screws	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Air filters	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	

w = weeks
m = months
a = years

1.3 Maintenance tasks

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	1 a	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).	X	W1024
QL1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL1	-	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.		W1009
QL1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect intercooler drain system.	X	W0502
				Inspect service indicator of air filter.		W0503
				Check relief bores of coolant pump(s).		W0505
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Drain off water and contamination from fuel prefilter.	X	W0507
				Check service indicator of fuel prefilter.	X	W0508
				Inspect battery-charging generator for contamination and clean if necessary.		W0525
QL1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL1	1000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.		W1241
QL1	1000	2 a	Primary fuel filter	Fit new primary fuel filter or primary fuel filter insert.		W1481
QL1	3000	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! First adjustment after 1,000 hours!.		W1207
QL1	3000	2 a	Coolant filters	Fit new coolant filters.		W1036
QL1	3000	2 a	Exhaust gas recirculation	Check operation of exhaust flap actuator.		W1551
QL3	4500	6 a	NOx sensor	Fit new sensor.		W1525
QL3	4500	6 a	Lambda sensor	Fit new sensor.		W1526
QL1	4500	9 a	Fuel injectors / injectors	Replace fuel injectors / injectors.		W1006
QL1	4500	9 a	Engine governor	Reset parameters of drift correction (CDC) and enter injector coding (IIG).		W1636
QL1	4500	6 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL1	4500	9 a	Engine mounts	Check securing screws of trunnion mounts for secure seating.		W1610
QL1	4500	18 a	Turbochargers	Check turbine housing externally (visual inspection) and internally (endoscopic examination) for cracks, replace if necessary (MTU ZR turbo-charger).		W1577
QL3	4500	6 a	Exhaust gas recirculation	Endoscopy EGR cooler, clean as needed.		W1524
				Replace flaps and actuators.		W1630
QL3	4500	5 a	Engine mounts	Check buffer clearance of resilient mounts Check proper seating of securing screws.		W1026
				Check for cracks. Measure cracks if necessary. Measure height of rubber elements.		W1462
QL3	9000	18 a	Turbochargers	Overhaul turbochargers.		W1038
QL3	9000	18 a	Component maintenance	Before starting maintenance work, drain the coolant and flush the cooling systems.		W2000
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.		W2001
				Clean air ducting.		W2002
				Clean intercooler and inspect for leakage.		W2003

w = weeks
m = months
a = years

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Qualifica- tion level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
				Fit new high-pressure fuel sensor.		W2004
				Check engine coolant thermostat and fit new thermal actuator.		W2006
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Inspect centrifugal oil filter for wear.		W2109
				Overhaul starter.		W2010
				Overhaul charge-air coolant pump.		W2070
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul engine coolant pump.		W2110
				Clean and leak-test low-pressure charge-air cooler.		W2153
QL3	9000	6 a	Crankcase breathers	Fit new fine separator for oil mist.		W1547
QL3	9000	18 a	Fuel delivery pump	Fit new fuel delivery pump.		W1051
QL3	9000	18 a	Battery charging generator	Overhaul battery-charging generator.		W1043
QL3	9000	18 a	Cylinder heads	Overhaul cylinder heads.		W1134
QL3	9000	6 a	Hose lines	Replace all hose lines.		W1251
QL4	9000	18 a	Extended component maintenance	Completely disassemble the engine. Inspect engine components as per assembly instructions and repair or fit new components as required.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new high-pressure fuel pump.		W3007
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Exhaust turbocharger: Inspect turbine (high-pressure) with borescope.		W3020
				Check gear train for cracks, replace bushings, check axles and replace if necessary.		W3063
				Replace fuel overpressure valve in distributor.		W3072
				Fit new engine oil pump.		W3073
				Replace swing followers and swing-follower shafts.		W3103
				Fit new vibration damper.		W3108
QL4	9000	18 a	Engine mounts	Fit new Trunnion mount element.		W1124
Transmission						
QL1	Daily	-	Transmission operation	Check gear oil level.		X4032
				Check general condition of gearbox and inspect for leaks.		X4033
				Check for abnormal running noise and vibration.		X4034
QL1	500	-	Gear oil filter	Replace transmission oil filter.		X4035
QL3	1000	-	Transmission mounts	Check resilient gearbox mounting (visual inspection).		X4022
QL1	1000	-	Screw connections	Check threaded joints for secure seating.		X4041
QL1	1000	-	Connectors/wiring harness	Check all electrical wiring components for security chafing or damage.		X4037
QL1	2000	6 m	Torsional resilient coupling	Check torsionally resilient coupling (visual inspection).		X4023
QL1	9000	-	Suction strainer (magnetic)	Clean magnetic suction strainer.		X4039
QL3	9000	-	Transmission	Forward transmission to manufacturer (Twin Disc) for overhaul.		X4080
Preheater / Jacket water heater						
QL1	-	1 a	Preheater / Jacket water heater	Overhaul preheater / jacket water heater.		X5601
QL1	-	5 a	Fuel pump	Replace pump.		X5603
QL1	18000	-	Circulation pump	Replace pump.		X5604
Frac Pack						
QL1	Daily	-	Exhaust silencer	Check condensate drain of silencer.	X	X8100
w = weeks m = months a = years						

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
QL1	Daily	-	Frac Pack securing screws	Check visually for damage.	X	X8101
QL1	500	-	Frac Pack securing screws	Check for secure seating.	X	X8102
QL1	500	2 a	Air filters	Fit new air filters.	X	X8103
w = weeks m = months a = years						